Funcții de mai multe variab

(1) Continuitatea funcției:

$$f(x,y)=\begin{cases}\dfrac{2(x-y)}{x+3y}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$$

$$\lim_{(x,y)\to(0,0)}\frac{2(x-y)}{x+3y}=\lim_{\substack{x\to0\\y\to0}}\frac{2(x-y)}{x+3y}=\lim_{\substack{x\to0\\y=mx}}\frac{2x-2mx}{2x+3mx}=$$

$$=\frac{2-2m}{2+3m}\Rightarrow \text{limita nu există, deoarece depinde de } m$$

$\Rightarrow$ funcția nu este continuă în origine $\Rightarrow$ f continuă pe $\mathbb{R}^2\setminus\{(0,0)\}$.

(2) Continuitatea funcției

$$f(x,y)=\begin{cases}\dfrac{x+y}{x^2+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$$

$$\lim_{(x,y)\to(0,0)}\frac{x+y}{x^2+y^2}=\lim_{\substack{x\to0\\y=mx}}\frac{x+mx}{x^2+m^2x^2}=\lim_{\substack{x\to0\\y=mx}}\frac{1+m}{x+m^2x}$$

Dar $\lim\limits_{\substack{x\searrow0\\y=mx}}\frac{1+m}{x(1+m^2)}\neq\lim\limits_{\substack{x\nearrow0\\y=mx}}\frac{1+m}{x(1+m^2)}\Rightarrow$ nu există limită

în origine $\Rightarrow$ f este continuă pe $\mathbb{R}^2\setminus\{(0,0)\}$

③ Continuitatea funcției f

$$f(x,y)=\begin{cases}\frac{1}{xy}\cdot\sin\frac{x^3y^2}{x^2+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$$

$$\lim_{(x,y)\to(0,0)} f(x,y)=\lim_{(x,y)\to(0,0)}\frac{1}{xy}\cdot\sin\frac{x^3y^2}{x^2+y^2}=\lim_{(x,y)\to(0,0)}\frac{\sin\left(\frac{x^3y^2}{x^2+y^2}\right)}{\frac{x^3y^2}{x^2+y^2}}\cdot\frac{x^3y^2}{(x^2+y^2)xy}$$

($\frac{\sin(\cdot)}{\cdot}\to 1$)

$$=\lim_{(x,y)\to(0,0)}\frac{x^2y}{x^2+y^2}$$

$$\left|\frac{x^2y}{x^2+y^2}\right|\spadesuit=\frac{x^2\cdot|y|}{x^2+y^2}\leq|y|\xrightarrow{y\to 0}0\Rightarrow$$

($\frac{x^2}{x^2+y^2}\leq 1$)

$$\lim_{(x,y)\to(0,0)} f(x,y)=0\Rightarrow f \text{ este continuă pe } \mathbb{R}^2.$$

④ Continuitatea funcției f

$$f(x,y)=\begin{cases}\sqrt{1-x^2-y^2}, & x^2+y^2\leq 1\\ 0, & x^2+y^2>1\end{cases}$$

$$\lim_{x^2+y^2\searrow 1} f(x,y)=0$$

$$\lim_{x^2+y^2\nearrow 1} f(x,y)=\lim_{x^2+y^2\nearrow 1}\sqrt{1-(x^2+y^2)}=0$$

$\Rightarrow$ f este continuă în toate punctele (x,y) pentru care $x^2+y^2=1$ și cum funcțiile de pe ramuri sunt cont. =)

f este continuă pe $\mathbb{R}^2$

⑤ Continuitatea funcției f

$$f(x,y)=\begin{cases}\frac{3xy^2}{2x^2+3y^4}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$$

$$\lim_{(x,y)\to(0,0)} f(x,y)=\lim_{\substack{x\to 0\\ y^2=mx}}\frac{3x\,mx}{2x^2+3m^2x^2}=\frac{3m}{2+3m^2}\Rightarrow$$

limita nu există, deoarece depinde de $m\Rightarrow f$ este continuă pe $\mathbb{R}^2\setminus\{(0,0)\}$

6. **Continuitatea funcției f**

$$f(x,y)=\begin{cases}\dfrac{(x+y)^2+y}{3x+12(x+y)^2+y}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$$

$$\lim_{(x,y)\to(0,0)} f(x,y)=\lim_{\substack{x\to 0\\ y=mx}}\frac{(x+mx)^2+mx}{3x+12(x+mx)^2+mx}=\lim_{\substack{x\to 0\\ y=mx}}\frac{(m+1)^2x^2+mx}{3x+12(m+1)^2x^2+mx}=$$

$$=\lim_{\substack{x\to 0\\ y=mx}}\frac{(m+1)^2x+m}{12(m+1)^2x+m+3}=\frac{m}{m+3}\Rightarrow$$ limita nu există, deoarece depinde de $m\Rightarrow f$ este continuă pe $\mathbb{R}^2\setminus\{(0,0)\}$.

7. **Continuitatea funcției f.**

$$f(x,y)=\begin{cases}\dfrac{x^2y^3}{x^2+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$$

$$\left|\frac{x^2y^3}{x^2+y^2}\right|=\underbrace{\frac{x^2}{x^2+y^2}}_{\leq 1}\cdot y^2\cdot|y|\leq y^2\cdot|y|\xrightarrow{y\to 0}0\Rightarrow$$

$$\lim_{(x,y)\to(0,0)} f(x,y)=\lim_{(x,y)\to(0,0)}\frac{x^2y^3}{x^2+y^2}=0\Rightarrow f \text{ este cont. în origine}\Rightarrow$$

f este continuă pe $\mathbb{R}^2$.

8. **Continuitatea funcției f.**

$$f(x,y)=\begin{cases}\dfrac{x^2y}{x^2+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$$

$$\left|\frac{x^2y}{x^2+y^2}\right|=\underbrace{\frac{x^2}{x^2+y^2}}_{\leq 1}\cdot|y|\leq|y|\xrightarrow{y\to 0}0\Rightarrow$$

$$\lim_{(x,y)\to(0,0)} f(x,y)=\lim_{(x,y)\to(0,0)}\frac{x^2y}{x^2+y^2}=0\Rightarrow f \text{ este continuă}$$

în origine și f este continuă pe $\mathbb{R}^2$

(9.) **Continuitatea funcției f**

$$f(x,y)=\begin{cases}\dfrac{xy}{x^2+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=0\end{cases}$$

$$\lim_{(x,y)\to(0,0)} f(x,y)=\lim_{\substack{x\to 0\\ y=mx}}\frac{x\cdot mx}{x^2+m^2x^2}=\lim_{\substack{x\to 0\\ y=mx}}\frac{mx^2}{x^2+m^2x^2}=\frac{m}{1+m^2}$$

$\Rightarrow$ limita nu există, deoarece depinde de m

$\Rightarrow$ f este continuă pe $\mathbb{R}^2\setminus\{(0,0)\}$

(10.) **Continuitatea funcției f**

$$f(x,y)=\begin{cases}\dfrac{x^2y}{x^4+y^2}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$$

$$\lim_{(x,y)\to(0,0)} f(x,y)=\lim_{\substack{x\to 0\\ y=mx^2}}\frac{x^2\cdot mx^2}{x^4+m^2x^4}=\lim_{\substack{x\to 0\\ y=mx^2}}\frac{mx^4}{x^4(1+m^2)}=\frac{m}{1+m^2}$$

$\Rightarrow$ limita nu există, deoarece ea depinde de m $\Rightarrow$

f e continuă pe $\mathbb{R}^2\setminus\{(0,0)\}$

(11.) **Continuitatea funcției f.**

$$f(x,y)=\begin{cases}(x^2+y^2)\sin\dfrac{1}{xy}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$$

$$\left|(x^2+y^2)\cdot\sin\frac{1}{xy}\right|\le|x^2+y^2|\xrightarrow[(x,y)\to(0,0)]{}0\Rightarrow$$

$\lim_{(x,y)\to(0,0)} f(x,y)=0\Rightarrow f$ este cont. în origine $\Rightarrow$ f cont. pe $\mathbb{R}^2$

(12) **Continuă parțial în origine, dar nu este continuă în acest punct**

$$f(x,y)=\begin{cases}\dfrac{xy^2+\sin(x^5+y^5)}{x^2+y^4}, & (x,y)\neq(0,0)\\ 0, & (x,y)=(0,0)\end{cases}$$

$$\lim_{(x,y)\to(0,0)} \frac{xy^2+\sin(x^3+y^5)}{x^2+y^4} = \lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2+y^4} + \lim_{(x,y)\to(0,0)} \underbrace{\frac{\sin(x^3+y^5)}{x^3+y^5}}_{\downarrow 1} \cdot \frac{x^3+y^5}{x^2+y^4}$$

$$= \lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2+y^4} + \lim_{(x,y)\to(0,0)} \frac{x^3+y^5}{x^2+y^4}$$

$$\left|\frac{x^3+y^5}{x^2+y^4}\right| = \frac{|x|\cdot x^2}{x^2+y^4} + \frac{|y|\cdot y^4}{x^2+y^4} \le |x|+|y| \xrightarrow{(x,y)\to(0,0)} 0 \Rightarrow$$

($\frac{x^2}{x^2+y^4} \le 1$, $\frac{y^4}{x^2+y^4} \le 1$)

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{(x,y)\to(0,0)} \frac{xy^2}{x^2+y^4} = \lim_{\substack{x\to 0 \\ y^2=mx}} \frac{mx^2}{x^2+m^2x^2} = \frac{m}{1+m^2} \Rightarrow$$

limita depinde de $m \Rightarrow f$ nu are limită în origine $\Rightarrow$ f nu este continuă în origine

(13.) Continuitatea funcției f.

$$f(x,y) = \begin{cases} \frac{x^2-y^2}{x^2+y^2}, & (x,y)\neq(0,0) \\ 0, & (x,y)=(0,0) \end{cases}$$

$$\lim_{(x,y)\to(0,0)} f(x,y) = \lim_{\substack{x\to 0 \\ y=ux}} \frac{x^2-u^2x^2}{x^2+u^2x^2} = \frac{1-u^2}{1+u^2} \Rightarrow \text{limita depinde de } u$$

$\Rightarrow f$ nu are limită în origine $\Rightarrow$ f este cont. pe $\mathbb{R}^2 \setminus \{(0,0)\}$

(14.) Continuitatea funcției f

$$f(x,y) = \begin{cases} \frac{1-\cos(x^3+y^3)}{x^2+y^2}, & (x,y)\neq(0,0) \\ 0, & (x,y)=(0,0) \end{cases}$$

$$\lim_{(x,y)\to(0,0)} \frac{1-\cos(x^3+y^3)}{x^2+y^2} = \lim_{(x,y)\to(0,0)} \frac{2\sin^2\frac{x^3+y^3}{2}}{x^2+y^2} =$$

$$= \lim_{(x,y)\to(0,0)} 2\cdot \frac{\sin^2\frac{x^3+y^3}{2}}{\left(\frac{x^3+y^3}{2}\right)^2} \cdot \left(\frac{x^3+y^3}{2}\right)^2 \cdot \frac{1}{x^2+y^2} = \lim_{(x,y)\to(0,0)} \frac{1}{2}\cdot\frac{(x^3+y^3)^2}{x^2+y^2}$$

$$\left|\frac{(x^3+y^3)^2}{x^2+y^2}\right| = \frac{[|x+y|\cdot|x^2-xy+y^2|]^2}{x^2+y^2} \le \frac{[|x+y|(|x^2+y^2|+|xy|)]^2}{|x^2+y^2|}$$

($|x^2-xy+y^2| \le 1$)

$|x+y| + \frac{|x+y|\cdot|xy|}{x^2+y^2} \leq \frac{|x+y|\cdot|x+y|}{x^2+y^2}$

$(x,y)\to(0,0)$

0

~~$(x+y)^2$~~.

$$\frac{(x^3+y^3)^2}{x^2+y^2} = \frac{|x^6+2x^3y^3+y^6|}{x^2+y^2} \leq \frac{x^6}{x^2+y^2} + \frac{2|x^3y^3|}{x^2+y^2} +$$

$$\frac{2y^6}{x^2+y^2} = \underbrace{\frac{x^2\cdot x^4}{(x^2+y^2)}}_{\leq 1} + \underbrace{\frac{2\,x^2\cdot|xy^3|}{x^2+y^2}}_{\leq 1} + \underbrace{\frac{2\,y^2\cdot y^4}{(x^2+y^2)}}_{\leq 1} \leq$$

$$x^4 + 2|xy^3| + 2y^4 \xrightarrow{(x,y)\to(0,0)} 0 \Rightarrow \lim_{(x,y)\to(0,0)} f(x,y) = 0 \Rightarrow$$

f este continuă în origine $\Rightarrow$ f este continuă pe $\mathbb{R}^2$.

(15.) **Continuitatea funcției f.**

$$f(x,y) = \begin{cases} \frac{xy + x^2y\ln|x+y|}{x^2+y^2}, & (x,y)\neq(0,0) \\ 0, & (x,y)=(0,0) \end{cases}$$

$$\left|\frac{x^2y\ln|x+y|}{x^2+y^2}\right| \leq \left|\frac{x^2y(x+y)}{x^2+y^2}\right| = \frac{|x^2y(x+y)|}{x^2+y^2} \leq \frac{|x^3y|}{x^2+y^2} + \frac{|x^2y^2|}{x^2+y^2}$$

$$= \underbrace{\frac{x^2|xy|}{x^2+y^2}}_{\leq 1} + \underbrace{\frac{x^2y^2}{x^2+y^2}}_{\leq 1} \leq |xy| + y^2 \longrightarrow 0$$

$$\Rightarrow \lim_{(x,y)\to(0,0)} f(x,y) = \lim_{(x,y)\to(0,0)} \frac{xy}{x^2+y^2} = \lim_{\substack{x\to 0 \\ y=ux}} \frac{ux^2}{x^2+u^2x^2} = \frac{u}{1+u^2} \Rightarrow$$

limita depinde de u $\Rightarrow$ limita nu există $\Rightarrow$ f nu este continuă în origine.

(16.) **Derivatele parțiale de ordinul întâi**

a) $f(x,y) = e^{x-y^2}$

$$\boxed{\frac{\partial f}{\partial x} = e^{x-y^2}} \qquad \boxed{\frac{\partial f}{\partial y} = -2y\cdot e^{x-y^2}}$$

b) $f(x,y,z) = e^{x^2+y^2} \cdot \sin^2 z$

$$\frac{\partial f}{\partial x} = 2x \cdot e^{x^2+y^2} \cdot \sin^2 z \qquad \frac{\partial f}{\partial z} = e^{x^2+y^2} \cdot 2 \sin z \cos z$$

$$\frac{\partial f}{\partial y} = 2y \cdot e^{x^2+y^2} \cdot \sin^2 z$$

c) $f(x,y) = xy \operatorname{arctg} \frac{x+y}{1-xy}, \quad xy \neq 1$

$$\frac{\partial f}{\partial x} = y \operatorname{arctg} \frac{x+y}{1-xy} + xy \cdot \frac{1}{1+\left(\frac{x+y}{1-xy}\right)^2} \cdot \frac{1-xy-(x+y)\cdot(-y)}{(1-xy)^2} =$$

$$= y \operatorname{arctg} \frac{x+y}{1-xy} + xy \cdot \frac{(1-xy)^2}{(1-xy)^2+(x+y)^2} \cdot \frac{1-2xy+xy+y^2}{(1-xy)^2} =$$

$$= y \operatorname{arctg} \frac{x+y}{1-xy} + xy \cdot \frac{1+y^2}{1-2xy+x^2y^2+x^2+2xy+y^2} = y \operatorname{arctg} \frac{x+y}{1-xy} + \frac{xy \cdot (1+y^2)}{1+x^2y^2+x^2+y^2}$$

$$\frac{\partial f}{\partial y} = x \operatorname{arctg} \frac{x+y}{1-xy} + xy \cdot \frac{1+x^2}{1+x^2y^2+x^2+y^2}$$

d) $f(x,y,z) = x^{y^z}$

$$\frac{\partial f}{\partial x} = x^{y^z-1} \cdot y^z \qquad \frac{\partial f}{\partial y} = x^{y^z} \cdot \ln x \cdot y^{z-1} \cdot z$$

$$\frac{\partial f}{\partial z} = x^{y^z} \cdot \ln x \cdot y^z \cdot \ln y$$

e) $f(x,y,z) = \ln(x^y y^z z^x), \quad x>0, y>0$

$$\frac{\partial f}{\partial x} = \frac{1}{x^y y^z z^x} \cdot (x^{y-1} \cdot y \cdot y^z \cdot z^x + x^y \cdot y^z \cdot z^x \cdot \ln z) =$$

$$= \frac{y}{x} + \ln z$$

$$\frac{\partial f}{\partial z} = \frac{1}{x^y \cdot y^z \cdot z^x} \left(x^y \cdot y^z \cdot \ln y \cdot z^x + x^y \cdot y^z \cdot z^{x-1} \cdot x \right) =$$

$$= \frac{x}{z} + \ln y$$

$$\frac{\partial f}{\partial y} = \frac{1}{x^y \cdot y^z \cdot z^x}\left(x^y \cdot \ln x \cdot y^z \cdot z^x + x^y \cdot y^{z-1} \cdot z \cdot z^x\right).$$

$$= \frac{z}{y} + \ln x$$

(17.) Derivatele partiale de ordinul doi

a) $f(x,y) = \ln(x+y^2)$ Derivate partiale de ord. 1

$$\frac{\partial f}{\partial x} = \frac{1}{x+y^2}, \quad \frac{\partial f}{\partial y} = \frac{2y}{x+y^2}$$

Derivate partiale de ordinul 2

$$\frac{\partial^2 f}{\partial x^2} = \frac{-1}{(x+y^2)^2} \qquad \frac{\partial^2 f}{\partial y^2} = \frac{2(x+y^2) - 2y \cdot 2y}{(x+y^2)^2} = \frac{2(x-y^2)}{(x+y^2)^2}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{-2y}{(x+y^2)^2}$$

b) $f(x,y) = x^3 + xy$

Derivatele partiale de ordinul 1:

$$\frac{\partial f}{\partial x} = 3x^2 + y \qquad \frac{\partial f}{\partial y} = x$$

Derivatele partiale de ordinul 2:

$$\frac{\partial^2 f}{\partial x^2} = 6x \qquad \frac{\partial^2 f}{\partial y^2} = 0 \qquad \frac{\partial^2 f}{\partial x \partial y} = 1$$

c) $f(x,y,z) = \ln(x+y+z)$

Derivatele partiale de ordinul 1

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial y} = \frac{\partial f}{\partial z} = \frac{1}{x+y+z}$$

Derivatele partiale de ordinul 2

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 f}{\partial z^2} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial z} = \frac{\partial^2 f}{\partial z \partial x} = \frac{-1}{(x+y+z)^2}$$

d) $f(x,y)=\frac{x-y}{x+y}$

Derivatele partiale de ordinul 1

$$\frac{\partial f}{\partial x}=\frac{x+y-(x-y)}{(x+y)^2}=\frac{2y}{(x+y)^2}$$

$$\frac{\partial f}{\partial y}=\frac{-(x+y)-(x-y)}{(x+y)^2}=\frac{-2x}{(x+y)^2}$$

Derivatele partiale de ordinul 2

$$\frac{\partial^2 f}{\partial x^2}=\frac{-4y}{(x+y)^3} \qquad \frac{\partial^2 f}{\partial y^2}=\frac{4x}{(x+y)^3}$$

$$\frac{\partial^2 f}{\partial x\partial y}=\frac{-2(x+y)^2+4x(x+y)}{(x+y)^4}=\frac{-2(x+y)+4x}{(x+y)^3}=\frac{2x-2y}{(x+y)^3}=\frac{2(x-y)}{(x+y)^3}$$

e) $f(x,y,z)=y\sin(x+z)$

Derivatele partiale de ordinul 1

$$\frac{\partial f}{\partial x}=y\cos(x+z) \qquad \frac{\partial f}{\partial y}=\sin(x+z) \qquad \frac{\partial f}{\partial z}=y\cos(x+z)$$

Derivatele partiale de ordinul 2.

$$\frac{\partial^2 f}{\partial x^2}=-y\sin(x+z) \qquad \frac{\partial^2 f}{\partial y^2}=0 \qquad \frac{\partial^2 f}{\partial z^2}=-y\sin(x+z)$$

$$\frac{\partial^2 f}{\partial x\partial y}=\cos(x+z) \qquad \frac{\partial^2 f}{\partial y\partial z}=\cos(x+z) \qquad \frac{\partial^2 f}{\partial z\partial x}=-y\sin(x+z).$$

(18.) **Operatorul lui Laplace / laplacianul funcției f**

$$\Delta f=\frac{\partial^2 f}{\partial x^2}+\frac{\partial^2 f}{\partial y^2}+\frac{\partial^2 f}{\partial z^2}$$

$$\Delta^* f=\left(\frac{\partial f}{\partial x}\right)^2+\left(\frac{\partial f}{\partial y}\right)^2+\left(\frac{\partial f}{\partial z}\right)^2$$

a) $f(x,y,z)=x^3+y^3+z^3-3xyz$

$$\frac{\partial f}{\partial x}=3x^2-3yz=3(x^2-yz) \qquad \frac{\partial f}{\partial y}=3y^2-3xz=3(y^2-xz)$$

$$\frac{\partial f}{\partial z}=3z^2-3xy=3(z^2-xy)$$

$$\Delta^* f=9\left[(x^2-yz)^2+(y^2-xz)^2+(z^2-xy)^2\right]$$

$$\frac{\partial^2 f}{\partial x^2}=6x \qquad \frac{\partial^2 f}{\partial y^2}=6y \qquad \frac{\partial^2 f}{\partial z^2}=6z \Rightarrow \Delta f=6(x+y+z)$$

b) $g(x,y,z) = \frac{1}{\sqrt{x^2+y^2+z^2}}$

$$\frac{\partial g}{\partial x} = \frac{-1}{x^2+y^2+z^2} \cdot \frac{2x}{2\sqrt{x^2+y^2+z^2}} = \frac{-x}{(x^2+y^2+z^2)^{3/2}}$$

$$\frac{\partial g}{\partial y} = \frac{-y}{(x^2+y^2+z^2)^{3/2}} \qquad \frac{\partial g}{\partial z} = \frac{-z}{(x^2+y^2+z^2)^{3/2}}$$

$$\Rightarrow \Delta^* g = \left(\frac{\partial g}{\partial x}\right)^2 + \left(\frac{\partial g}{\partial y}\right)^2 + \left(\frac{\partial g}{\partial z}\right)^2 = \frac{x^2+y^2+z^2}{(x^2+y^2+z^2)^3} = \frac{1}{(x^2+y^2+z^2)^2}$$

$$\frac{\partial^2 g}{\partial x^2} = \frac{-(x^2+y^2+z^2)^{3/2} + 2x^2 \cdot \frac{3}{2}(x^2+y^2+z^2)^{1/2}}{(x^2+y^2+z^2)^3} = \frac{-(x^2+y^2+z^2)+3x^2}{(x^2+y^2+z^2)^{5/2}} =$$

$$= \frac{2x^2-y^2-z^2}{(x^2+y^2+z^2)^{5/2}}$$

$$\frac{\partial^2 g}{\partial y^2} = \frac{2y^2-x^2-z^2}{(x^2+y^2+z^2)^{5/2}} \qquad \frac{\partial^2 g}{\partial z^2} = \frac{2z^2-x^2-y^2}{(x^2+y^2+z^2)^{5/2}}$$

$$\Rightarrow \Delta g = \frac{-2(x^2+y^2+z^2)+2x^2+2y^2+2z^2}{(x^2+y^2+z^2)^{5/2}} = 0.$$

(19.) $\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \quad f(x,y) = \varphi(u(x,y), v(x,y))$

a) $\begin{cases} u(x,y) = x+y \\ v(x,y) = x^2+y^2 \end{cases}$

$$\frac{\partial u}{\partial x} = 1 \quad \frac{\partial u}{\partial y} = 1 \quad \frac{\partial v}{\partial x} = 2x \quad \frac{\partial v}{\partial y} = 2y$$

$$\Rightarrow \frac{\partial f}{\partial x} = \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial \varphi}{\partial v} \cdot \frac{\partial v}{\partial x} = \frac{\partial \varphi}{\partial u} + 2x \frac{\partial \varphi}{\partial v}$$

$$\frac{\partial f}{\partial y} = \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial \varphi}{\partial v} \cdot \frac{\partial v}{\partial y} = \frac{\partial \varphi}{\partial u} + 2y \frac{\partial \varphi}{\partial v}$$

b) $\begin{cases} u(x,y) = x^2-y^2 \\ v(x,y) = e^{xy} \end{cases}$

$$\frac{\partial u}{\partial x} = 2x \quad \frac{\partial u}{\partial y} = -2y \quad \frac{\partial v}{\partial x} = x \cdot e^{xy} \quad \frac{\partial v}{\partial y} = y \cdot e^{xy}$$

$$\frac{\partial f}{\partial x} = \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial \varphi}{\partial v} \cdot \frac{\partial v}{\partial x} = 2x \frac{\partial \varphi}{\partial u} + y \cdot e^{xy} \cdot \frac{\partial \varphi}{\partial v}$$

$$\frac{\partial f}{\partial y} = \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial \varphi}{\partial v} \cdot \frac{\partial v}{\partial y} = -2y \frac{\partial \varphi}{\partial u} + x \cdot e^{xy} \cdot \frac{\partial \varphi}{\partial v}$$

(20) $\frac{\partial f}{\partial x}$, $\frac{\partial f}{\partial y}$, $f(u,v) = \operatorname{arctg}\frac{u}{v}$

$$\begin{cases} u = x\sin y \\ v = x\cos y \end{cases}$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x}$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial y}$$

$$\frac{\partial u}{\partial x} = \sin y \,,\quad \frac{\partial u}{\partial y} = x\cos y \,,\quad \frac{\partial v}{\partial x} = \cos y \,,\quad \frac{\partial v}{\partial y} = -x\sin y$$

$$\frac{\partial f}{\partial u} = \frac{1}{1+\left(\frac{u}{v}\right)^2}\cdot\frac{1}{v} = \frac{v^2}{u^2+v^2}\cdot\frac{1}{v} = \frac{v}{u^2+v^2}$$

$$\frac{\partial f}{\partial v} = \frac{1}{1+\left(\frac{u}{v}\right)^2}\cdot\frac{-u}{v^2} = \frac{-u}{u^2+v^2}$$

$$\Rightarrow \frac{\partial f}{\partial x} = \frac{v}{u^2+v^2}\cdot\sin y - \frac{u}{u^2+v^2}\cdot\cos y = \frac{x\sin y\cos y - x\sin y\cos y}{u^2+v^2} = 0$$

$$\frac{\partial f}{\partial y} = \frac{v}{u^2+v^2}\cdot x\cos y + \frac{u}{u^2+v^2}\cdot x\sin y = \frac{x^2\cos^2 y + x^2\sin^2 y}{u^2+v^2} =$$

$$= \frac{x^2}{u^2+v^2} = \frac{x^2}{x^2\sin^2 x + x^2\cos^2 y} = 1$$

(21) $f = f(u,v)$, $u = xy$, $v = \frac{x}{y}$; $\frac{\partial^2 f}{\partial x^2} = ?$, $\frac{\partial^2 f}{\partial x \partial y} = ?$, $\frac{\partial^2 f}{\partial y^2} = ?$

$$\frac{\partial u}{\partial x} = y \qquad \frac{\partial v}{\partial x} = \frac{1}{y}$$

$$\frac{\partial u}{\partial y} = x \qquad \frac{\partial v}{\partial y} = \frac{-x}{y^2}$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x} = y\cdot\frac{\partial f}{\partial u} + \frac{1}{y}\cdot\frac{\partial f}{\partial v}$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial y} = x\frac{\partial f}{\partial u} - \frac{x}{y^2}\cdot\frac{\partial f}{\partial v}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}\left(y\cdot\frac{\partial f}{\partial u} + \frac{1}{y}\cdot\frac{\partial f}{\partial v}\right) =$$

$$= y\cdot\frac{\partial^2 f}{\partial u^2}\cdot\frac{\partial u}{\partial x} + y\cdot\frac{\partial^2 f}{\partial v\partial u}\cdot\frac{\partial v}{\partial x} + \frac{1}{y}\frac{\partial^2 f}{\partial u\partial v}\frac{\partial u}{x} +$$

$$\frac{1}{y}\cdot\frac{\partial^2 f}{\partial v^2}\cdot\frac{\partial v}{\partial x} = y^2\frac{\partial^2 f}{\partial u^2} + 2\frac{\partial^2 f}{\partial u\partial v} + \frac{1}{y^2}\frac{\partial^2 f}{\partial v^2} =$$

$$= \frac{u}{v}\frac{\partial^2 f}{\partial u^2} + 2\cdot\frac{\partial^2 f}{\partial u\partial v} + \frac{v}{u}\cdot\frac{\partial^2 f}{\partial v^2}$$

$$\frac{\partial^2 f}{\partial x\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial u}\cdot x + \frac{\partial f}{\partial v}\cdot\frac{-x}{y^2}\right) =$$

$$= \frac{\partial}{\partial x}\left(x\left(\frac{\partial f}{\partial u} - \frac{1}{y^2}\frac{\partial f}{\partial v}\right)\right) = \frac{\partial f}{\partial u} - \frac{1}{y^2}\cdot\frac{\partial f}{\partial v} + x\left(\frac{\partial^2 f}{\partial u^2}\cdot\frac{\partial u}{\partial x} +$$

$$\frac{\partial^2 f}{\partial v\partial u}\cdot\frac{\partial v}{\partial x} - \frac{1}{y^2}\frac{\partial^2 f}{\partial u\partial v}\cdot\frac{\partial u}{\partial x} - \frac{1}{y^2}\cdot\frac{\partial^2 f}{\partial v^2}\cdot\frac{\partial v}{\partial x}\right) = \frac{\partial f}{\partial u} -$$

$$\frac{1}{y^2}\cdot\frac{\partial f}{\partial v} + x\left(\frac{\partial^2 f}{\partial u^2}\cdot y + \frac{\partial^2 f}{\partial u\partial v}\cdot\frac{1}{y} - \frac{\partial^2 f}{\partial u\partial v}\cdot\frac{1}{y} - \frac{1}{y^3}\cdot\frac{\partial^2 f}{\partial v^2}\right) =$$

$$= xy\cdot\frac{\partial^2 f}{\partial u^2} - \frac{x}{y^3}\frac{\partial^2 f}{\partial v^2} + \frac{\partial f}{\partial u} - \frac{1}{y^2}\frac{\partial f}{\partial v} = u\cdot\frac{\partial^2 f}{\partial u^2} - \frac{v^2}{u}\cdot\frac{\partial^2 f}{\partial v^2} +$$

$$\frac{\partial f}{\partial u} - \frac{v}{u}\cdot\frac{\partial f}{\partial v}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial u}\cdot x + \frac{\partial f}{\partial v}\cdot\frac{-x}{y^2}\right) =$$

$$= x\cdot\frac{\partial^2 f}{\partial u^2}\cdot\frac{\partial u}{\partial y} + x\cdot\frac{\partial^2 f}{\partial v\partial u}\cdot\frac{\partial v}{\partial y} + \frac{2x}{y^3}\cdot\frac{\partial f}{\partial v} - \frac{x}{y^2}\frac{\partial^2 f}{\partial u\partial v}\cdot\frac{\partial u}{\partial y} -$$

$$\frac{x}{y^2}\cdot\frac{\partial^2 f}{\partial v^2}\cdot\frac{\partial v}{\partial y} = x^2\frac{\partial^2 f}{\partial u^2} - \frac{x^2}{y^2}\cdot\frac{\partial^2 f}{\partial u\partial v} + \frac{2x}{y^3}\frac{\partial f}{\partial v} -$$

$$\frac{x^2}{y^2}\frac{\partial^2 f}{\partial u\partial v} + \frac{x^2}{y^4}\cdot\frac{\partial^2 f}{\partial v^2} = uv\frac{\partial^2 f}{\partial u^2} - 2v^2\frac{\partial^2 f}{\partial u\partial v} + \frac{v^3}{u}\cdot\frac{\partial^2 f}{\partial v^2} + \frac{2v^2}{u}\cdot\frac{\partial f}{\partial v}$$

(22) $E = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial x\partial y} + \frac{\partial^2 f}{\partial y^2}$

$f(u,v) = \ln(u^2+v^2)$ $\quad u = xy$ $\quad v = x^2 - y^2$

$\frac{\partial u}{\partial x} = y$ $\quad \frac{\partial u}{\partial y} = x$ $\quad \frac{\partial v}{\partial x} = 2x$ $\quad \frac{\partial v}{\partial y} = -2y$

$\frac{\partial f}{\partial u} = \frac{2u}{u^2+v^2}$ $\quad \frac{\partial f}{\partial v} = \frac{2v}{u^2+v^2}$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x} = \frac{2u}{u^2+v^2}\cdot y + \frac{2v}{u^2+v^2}\cdot 2x = \frac{2xy^2}{(x^2-y^2)^2+x^2y^2} +$$

$$\frac{4x(x^2-y^2)}{x^2y^2+(x^2-y^2)^2} = \frac{4x^3-2xy^2}{x^4+y^4-x^2y^2}$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial y} = \frac{2xy}{x^4+y^4-x^2y^2}\cdot x - \frac{2(x^2-y^2)}{x^4+y^4-x^2y^2}\cdot 2y =$$

$$= \frac{2x^2y - 4x^2y + 4y^3}{x^4+y^4-x^2y^2} = \frac{4y^3-2x^2y}{x^4+y^4-x^2y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{-2(2x^6 - y^2x^4 - 5x^4x^2 + y^6)}{(x^4+y^4-x^2y^2)^2}$$

$$\frac{\partial^2 f}{\partial x\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{4yx(x^4 - 4y^2x^2 + y^4)}{(x^4-y^2x^2+y^4)^2}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = -\frac{2(2y^6 - x^2y^4 - 5x^4y^2 + x^6)}{(x^4+y^4-x^2y^2)^2}$$

$$E = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial x\partial y} + \frac{\partial^2 f}{\partial y^2} = \frac{-6(x^6 - 2x^4y^2 - 2x^2y^4 + y^6) + 4xy(x^4 - 4y^2x^2 + y^4)}{(x^4+y^4-x^2y^2)^2}$$

(23.) $f = f(u,v)$

$$\begin{cases} u(x,y) = x^2+y^2 \\ v(x,y) = xy \end{cases}$$

$$\frac{\partial u}{\partial x} = 2x \quad \frac{\partial u}{\partial y} = 2y \quad \frac{\partial v}{\partial x} = y \quad \frac{\partial v}{\partial y} = x$$

$$\frac{\partial f}{\partial x} = \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial x} = 2x\frac{\partial f}{\partial u} + y\cdot\frac{\partial f}{\partial v}$$

$$\frac{\partial f}{\partial y} = \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial y} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial y} = 2y\cdot\frac{\partial f}{\partial u} + x\cdot\frac{\partial f}{\partial v}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}\left(2x\cdot\frac{\partial f}{\partial u} + y\cdot\frac{\partial f}{\partial v}\right) = 2\frac{\partial f}{\partial u} + 2x\frac{\partial^2 f}{\partial u^2}\cdot\frac{\partial u}{\partial x} +$$

$$2x\frac{\partial^2 f}{\partial v\partial u}\cdot\frac{\partial v}{\partial x} + y\cdot\frac{\partial^2 f}{\partial v^2}\cdot\frac{\partial v}{\partial x} + y\cdot\frac{\partial^2 f}{\partial u\partial v}\cdot\frac{\partial u}{\partial x} = 4x^2\frac{\partial^2 f}{\partial u^2} + 4xy\frac{\partial^2 f}{\partial u\partial v} +$$

$$y^2\frac{\partial^2 f}{\partial v^2} + 2\frac{\partial f}{\partial u}$$

$$\frac{\partial^2 f}{\partial x\partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}\left(2y\frac{\partial f}{\partial u} + x\frac{\partial f}{\partial v}\right) = 2y\frac{\partial^2 f}{\partial u^2}\cdot\frac{\partial u}{\partial x} +$$

$$2y\cdot\frac{\partial^2 f}{\partial v\partial u}\cdot\frac{\partial v}{\partial x} + \frac{\partial f}{\partial v} + x\cdot\frac{\partial^2 f}{\partial u\partial v}\cdot\frac{\partial u}{\partial x} + x\frac{\partial^2 f}{\partial v^2}\cdot\frac{\partial v}{\partial x} =$$

$$= 4xy\cdot\frac{\partial^2 f}{\partial u^2} + \frac{\partial^2 f}{\partial u\partial v}(2y^2+x^2) + xy\frac{\partial^2 f}{\partial v^2} + \frac{\partial f}{\partial v} = 4v\cdot\frac{\partial^2 f}{\partial u^2} +$$

$$2u\cdot\frac{\partial^2 f}{\partial u\partial v} + v\cdot\frac{\partial^2 f}{\partial v^2} + \frac{\partial f}{\partial v}$$

$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}\left(2y \cdot \frac{\partial f}{\partial u} + x \frac{\partial f}{\partial v}\right) = 2\frac{\partial f}{\partial u} + 2y \frac{\partial^2 f}{\partial u^2} \cdot \frac{\partial u}{\partial y} +$

$2y \cdot \frac{\partial^2 f}{\partial v \partial u} \cdot \frac{\partial v}{\partial y} + x \cdot \frac{\partial^2 f}{\partial u \partial v} \cdot \frac{\partial u}{\partial y} + x \cdot \frac{\partial^2 f}{\partial v^2} \cdot \frac{\partial v}{\partial y} - 2\frac{\partial f}{\partial u} +$

$4y^2 \frac{\partial^2 f}{\partial u^2} + 4xy \frac{\partial^2 f}{\partial v \partial u} + x^2 \frac{\partial^2 f}{\partial v^2}$

(24.) $f(x,y) = \varphi(x-ay) + \psi(x+ay)$

φ, ψ admit derivate de ord 2.

Ec. coardei vibrante $\frac{\partial^2 f}{\partial y^2} = a^2 \cdot \frac{\partial^2 f}{\partial x^2}$

$\frac{\partial u}{\partial x} = 1 \quad \frac{\partial u}{\partial y} = -a$

$\frac{\partial v}{\partial x} = 1 \quad \frac{\partial v}{\partial y} = a$

Notăm $u(x,y) = x-ay$, $v(x,y) = x+ay$

$\frac{\partial f}{\partial x} = \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial \psi}{\partial v} \cdot \frac{\partial v}{\partial x} = \frac{\partial \varphi}{\partial u} + \frac{\partial \psi}{\partial v}$

$\frac{\partial f}{\partial y} = \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial \psi}{\partial v} \cdot \frac{\partial v}{\partial y} = -a \cdot \frac{\partial \varphi}{\partial u} + a \cdot \frac{\partial \psi}{\partial v}$

$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 \varphi}{\partial u^2} \cdot \frac{\partial u}{\partial x} + \frac{\partial^2 \psi}{\partial v^2} \cdot \frac{\partial v}{\partial x} = \frac{\partial^2 \varphi}{\partial u^2} + \frac{\partial^2 \psi}{\partial v^2}$

$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 \varphi}{\partial u^2} \cdot \frac{\partial u}{\partial y} \cdot (-a) + \frac{\partial^2 \psi}{\partial v^2} \cdot \frac{\partial v}{\partial y} \cdot a = a^2 \cdot \frac{\partial^2 \varphi}{\partial u^2} + a^2 \cdot \frac{\partial^2 \psi}{\partial v^2}$

$= a^2 \cdot \frac{\partial^2 f}{\partial x^2} \Rightarrow$ ec. coardei vibrante este satisfăcută.

(25.)

a) $f(x,y) = y\varphi(x^2-y^2) \quad \frac{\partial u}{\partial x} = 2x \quad \frac{\partial u}{\partial y} = -2y$

$u(x,y) = x^2-y^2$

$\frac{\partial f}{\partial x} = y \cdot \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial x} = 2xy \cdot \frac{\partial \varphi}{\partial u}$

$\frac{\partial f}{\partial y} = \varphi(x^2-y^2) + y \cdot \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial y} = \varphi(x^2-y^2) + -2y^2 \cdot \frac{\partial \varphi}{\partial u}$ $\Big\} \Rightarrow$

$\frac{1}{x} \cdot \frac{\partial f}{\partial x} + \frac{1}{y} \cdot \frac{\partial f}{\partial y} = 2y \frac{\partial \varphi}{\partial u} + \frac{1}{y}\varphi(x^2-y^2) - 2y \frac{\partial \varphi}{\partial u} = \frac{1}{y} \cdot \varphi(x^2-y^2) = \frac{1}{y^2} \cdot f$

b) $f(x,y) = e^y \cdot \varphi\left(y \cdot e^{\frac{x^2}{2y^2}}\right)$

$u = y \cdot e^{\frac{x^2}{2y^2}}$

$$\begin{cases} \frac{\partial u}{\partial x} = \frac{x \cdot e^{\frac{x^2}{2y^2}}}{y} \\ \frac{\partial u}{\partial y} = \frac{(y^2 - x^2) \cdot e^{\frac{x^2}{2y^2}}}{y^2} \end{cases}$$

$$\frac{\partial f}{\partial x} = \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial x} = \frac{\partial \varphi}{\partial u} \cdot \frac{x \cdot e^{\frac{x^2}{2y^2}}}{y}$$

$$\frac{\partial f}{\partial y} = e^y \cdot \varphi\left(y \cdot e^{\frac{x^2}{2y^2}}\right) + e^y \cdot \frac{\partial \varphi}{\partial u} \cdot \frac{(y^2 - x^2) \cdot e^{\frac{x^2}{2y^2}}}{y^2}$$

$$\Rightarrow (x^2 - y^2)\frac{\partial f}{\partial x} - xy\frac{\partial f}{\partial y} = \frac{\partial \varphi}{\partial y} \cdot \frac{x(x^2 - y^2) \cdot e^{\frac{x^2}{2y^2}}}{y} + \frac{x(y^2 - x^2) \cdot e^{\frac{x^2}{2y^2}}}{y} \cdot \frac{\partial \varphi}{\partial y}$$

$$+ xy \cdot \underbrace{e^y \cdot \varphi\left(y \cdot e^{\frac{x^2}{2y^2}}\right)}_{f} = xyf$$

c) $f(x,y) = \varphi(xy) + \sqrt{xy} \cdot \psi\left(\frac{y}{x}\right)$

Fie $u(x,y) = xy$, $v(x,y) = \frac{y}{x}$

$$\frac{\partial u}{\partial x} = y, \quad \frac{\partial u}{\partial y} = x, \quad \frac{\partial v}{\partial x} = \frac{-y}{x^2}, \quad \frac{\partial v}{\partial y} = \frac{1}{x}$$

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}\left(\varphi(xy) + \sqrt{xy} \cdot \psi\left(\frac{y}{x}\right)\right) = \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial x} + \sqrt{y} \cdot \frac{1}{2\sqrt{x}} \cdot \psi\left(\frac{y}{x}\right) +$$

$$\sqrt{xy}\,\frac{\partial \psi}{\partial v} \cdot \frac{\partial v}{\partial x} = \frac{\partial \varphi}{\partial u} \cdot y + \frac{1}{2}\sqrt{\frac{y}{x}} \cdot \psi\left(\frac{y}{x}\right) + \sqrt{xy} \cdot \frac{\partial \psi}{\partial v} \cdot \frac{-y}{x^2}$$

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}\left(\varphi(x,y) + \sqrt{xy} \cdot \psi\left(\frac{y}{x}\right)\right) = \frac{\partial \varphi}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{1}{2}\sqrt{\frac{x}{y}} \cdot \psi\left(\frac{y}{x}\right) +$$

$$\sqrt{xy}\,\frac{\partial \psi}{\partial v} \cdot \frac{\partial v}{\partial y} = \frac{\partial \varphi}{\partial u} \cdot x + \frac{1}{2}\sqrt{\frac{x}{y}} \cdot \psi\left(\frac{y}{x}\right) + \sqrt{xy} \cdot \frac{1}{x} \cdot \frac{\partial \psi}{\partial v}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial}{\partial x}\left(\frac{\partial \varphi}{\partial u} \cdot y + \frac{1}{2}\sqrt{\frac{y}{x}}\psi\left(\frac{y}{x}\right) + \sqrt{xy} \cdot \frac{\partial \psi}{\partial v} \cdot \frac{-y}{x^2}\right) =$$

$$= \frac{\partial^2 \varphi}{\partial u^2} \cdot y^2 - \frac{\sqrt{y}}{4\sqrt{x^3}}\psi\left(\frac{y}{x}\right) - \frac{1}{2}\sqrt{\frac{y}{x}}\,\frac{y}{x^2}\frac{\partial \psi}{\partial v} + \frac{y}{2}\sqrt{\frac{y}{x}} \cdot \frac{y}{x^2} \cdot \frac{\partial \psi}{\partial v} +$$

$$\frac{\partial^2 \psi}{\partial v^2} \cdot \frac{\sqrt{xy} \cdot y^2}{x^4}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial y}\left(\frac{\partial \varphi}{\partial u} \cdot x + \frac{1}{2}\sqrt{\frac{x}{y}}\psi\left(\frac{y}{x}\right) + \sqrt{xy} \cdot \frac{1}{x}\frac{\partial \psi}{\partial v}\right) =$$

$$= \frac{\partial^2 \varphi}{\partial u^2} \cdot x^2 - \frac{\sqrt{x}}{4\sqrt{y^3}}\psi\left(\frac{y}{x}\right) + \frac{1}{2}\sqrt{\frac{x}{y}}\frac{\partial \psi}{\partial v} \cdot \frac{1}{x} + \frac{1}{2} \cdot \frac{1}{\sqrt{xy}} \cdot \frac{\partial \psi}{\partial v} +$$

$$\sqrt{\frac{x}{y}} \cdot \frac{\partial^2 \psi}{\partial v^2} \cdot \frac{1}{x}$$

$$\Rightarrow x^2\frac{\partial^2 f}{\partial x^2} - y^2\frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 \varphi}{\partial u^2}x^2y^2 - \frac{1}{4}\sqrt{xy}\,\psi\left(\frac{y}{x}\right) + \sqrt{\frac{y}{x}} \cdot y\frac{\partial \psi}{\partial v} +$$

$$\frac{\partial^2 \psi}{\partial v^2}\sqrt{xy} \cdot \frac{y^2}{x^2} - x^2y^2 \cdot \frac{\partial^2 \varphi}{\partial u^2} + \frac{1}{4}\sqrt{xy}\,\psi\left(\frac{y}{x}\right) - \frac{1}{2}\sqrt{\frac{y}{x}}\,y\frac{\partial \psi}{\partial v} - \frac{1}{2}\sqrt{\frac{y}{x}}\,y\frac{\partial \psi}{\partial v}$$

$\sqrt{xy}\,\frac{\partial^2\varphi}{\partial v^2}\cdot\frac{y^2}{x^2}=0$

d) $f(x,y)=xy\,\varphi(\underbrace{x^2-y^2}_{u(x,y)})$

$\frac{\partial u}{\partial x}=2x \quad \frac{\partial u}{\partial y}=-2y$

$\frac{\partial f}{\partial x}=y\cdot\varphi(x^2-y^2)+xy\cdot\frac{\partial\varphi}{\partial u}\cdot\frac{\partial u}{\partial x}=y\cdot\varphi(x^2-y^2)+\frac{\partial\varphi}{\partial u}\cdot 2x^2y$

$\frac{\partial f}{\partial y}=x\cdot\varphi(x^2-y^2)+xy\cdot\frac{\partial\varphi}{\partial u}\cdot\frac{\partial u}{\partial y}=y\cdot\varphi(x^2-y^2)-\frac{\partial\varphi}{\partial u}\cdot 2xy^2$

$\Rightarrow xy^2\frac{\partial f}{\partial x}+x^2y\cdot\frac{\partial f}{\partial y}=y^2\cdot\underbrace{xy\varphi(x^2+y^2)}_{f}+\frac{\partial\varphi}{\partial u}\cdot 2x^3y^3+x^2\cdot\underbrace{xy\cdot\varphi(x^2+y^2)}_{f}$

$-\frac{\partial\varphi}{\partial u}\cdot 2x^3y^3=(x^2+y^2)\cdot f$

(26.) $f(x,y)=\varphi(\underbrace{\frac{y}{x}}_{u})+x\psi(\frac{y}{x}),\ x\neq 0$

$\frac{\partial u}{\partial x}=-\frac{y}{x^2} \quad \frac{\partial u}{\partial y}=\frac{1}{x}$

$\frac{\partial f}{\partial x}=\frac{\partial}{\partial x}\left(\varphi(\frac{y}{x})+x\psi(\frac{y}{x})\right)=\frac{\partial\varphi}{\partial u}\cdot\frac{\partial u}{\partial x}+\psi(\frac{y}{x})+x\cdot\frac{\partial\psi}{\partial u}\cdot\frac{\partial u}{\partial x}=$

$=-\frac{y}{x^2}\cdot\frac{\partial\varphi}{\partial u}+\psi(\frac{y}{x})-\frac{y}{x}\cdot\frac{\partial\psi}{\partial u}$

$\frac{\partial f}{\partial y}=\frac{\partial\varphi}{\partial u}\cdot\frac{\partial u}{\partial y}+x\cdot\frac{\partial\psi}{\partial u}\cdot\frac{\partial u}{\partial y}=\frac{\partial\varphi}{\partial u}\cdot\frac{1}{x}+\frac{\partial\psi}{\partial u}$

Derivatele partiale de ord. 2.

$\frac{\partial^2 f}{\partial x^2}=\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right)=\frac{\partial}{\partial x}\left(-\frac{y}{x^2}\cdot\frac{\partial\varphi}{\partial u}+\psi(\frac{y}{x})-\frac{y}{x}\cdot\frac{\partial\psi}{\partial u}\right)=$

$\frac{2y}{x^3}\cdot\frac{\partial\varphi}{\partial u}-\frac{y}{x^2}\cdot\frac{\partial\psi}{\partial u}-\frac{y}{x^2}\cdot\frac{\partial^2\varphi}{\partial u^2}\cdot\frac{\partial u}{\partial x}-\frac{\partial\psi}{\partial u}\cdot\frac{\partial u}{\partial x}-\frac{y}{x}\cdot\frac{\partial^2\psi}{\partial u^2}\cdot\frac{\partial u}{\partial x}=$

$=\frac{y}{2x^3}\cdot\frac{\partial\varphi}{\partial u}-\frac{y}{x^2}\cdot\frac{\partial\psi}{\partial u}+\frac{y^2}{x^4}\cdot\frac{\partial^2\varphi}{\partial u^2}+\frac{\partial\psi}{\partial u}\cdot$

$\frac{y}{x^2}+\frac{y^2}{x^3}\cdot\frac{\partial^2\psi}{\partial u^2}=\frac{2y}{x^3}\cdot\frac{\partial\varphi}{\partial u}+\frac{y^2}{x^4}\cdot\frac{\partial^2\varphi}{\partial u^2}+\frac{y^2}{x^3}\cdot\frac{\partial^2\psi}{\partial u^2}$

$\frac{\partial^2 f}{\partial x\partial y}=\frac{\partial}{\partial x}\left(\frac{\partial\varphi}{\partial u}\cdot\frac{1}{x}+\frac{\partial\psi}{\partial u}\right)=-\frac{1}{x^2}\cdot\frac{\partial\varphi}{\partial u}-\frac{\partial^2\varphi}{\partial u^2}\cdot\frac{y}{x^3}-$

$\frac{\partial^2\psi}{\partial u^2}\cdot\frac{y}{x^2}$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial \varphi}{\partial u}\cdot\frac{1}{x} + \frac{\partial \psi}{\partial u}\right) = \frac{\partial^2 \varphi}{\partial u^2}\cdot\frac{1}{x}\cdot\frac{\partial u}{\partial y} + \frac{\partial^2 \psi}{\partial u^2}\cdot\frac{\partial u}{\partial y} = \frac{1}{x^2}\cdot\frac{\partial^2 \varphi}{\partial u^2} + \frac{1}{x}\cdot\frac{\partial^2 \psi}{\partial u^2} \Rightarrow$$

$$x^2\cdot\frac{\partial^2 f}{\partial x^2} + 2xy\frac{\partial^2 f}{\partial x\,\partial y} + y^2\cdot\frac{\partial^2 f}{\partial y^2} + x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} - x\psi\left(\frac{y}{x}\right) =$$

$$2\cdot\frac{y}{x}\cdot\frac{\partial \varphi}{\partial u} + \frac{y^2}{x^2}\cdot\frac{\partial^2 \varphi}{\partial u^2} + \frac{y^2}{x}\cdot\frac{\partial^2 \psi}{\partial u^2} - 2\frac{y}{x}\cdot\frac{\partial \varphi}{\partial u} - 2\frac{y^2}{x^2}\cdot\frac{\partial^2 \varphi}{\partial u^2} -$$

$$2\frac{y^2}{x}\cdot\frac{\partial^2 \psi}{\partial u^2} + \frac{y^2}{x^2}\cdot\frac{\partial^2 \varphi}{\partial u^2} + \frac{y^2}{x}\cdot\frac{\partial^2 \psi}{\partial u^2} - \frac{y}{x}\cdot\frac{\partial \varphi}{\partial u} + x\cdot\psi\left(\frac{y}{x}\right) -$$

$$y\cdot\frac{\partial \psi}{\partial u} + \frac{\partial \varphi}{\partial u}\cdot\frac{y}{x} + y\cdot\frac{\partial \psi}{\partial u} - x\psi\left(\frac{y}{x}\right) = 0$$

(27.) Diferențialele de ord. I și II

$f(x,y,z) = \cos(x+2y+3z)$

$$\left.\begin{aligned}\frac{\partial f}{\partial x} &= -\sin(x+2y+3z)\\ \frac{\partial f}{\partial y} &= -2\sin(x+2y+3z)\\ \frac{\partial f}{\partial z} &= -3\sin(x+2y+3z)\end{aligned}\right\} \Rightarrow df = -\sin(x+2y+3z)(dx+2dy+3dz)$$

$$\frac{\partial^2 f}{\partial x^2} = -\cos(x+2y+3z) \qquad \frac{\partial^2 f}{\partial x\,\partial y} = -2\cos(x+2y+3z)$$

$$\frac{\partial^2 f}{\partial y^2} = -4\cos(x+2y+3z) \qquad \frac{\partial^2 f}{\partial y\,\partial z} = -6\cos(x+2y+3z)$$

$$\frac{\partial^2 f}{\partial z^2} = -9\cos(x+2y+3z) \qquad \frac{\partial^2 f}{\partial z\,\partial x} = -3\cos(x+2y+3z)$$

$$\Rightarrow d^2 f = -\cos(x+2y+3z)(dx^2+4dy^2+9dz^2+4dx\,dy+12dy\,dz+6dz\,dx) =$$

$$= -\cos(x+2y+3z)\cdot(dx+2dy+3dz)^2$$

(28.) Diferențialele de ord. I și II pentru funcțiile compuse

a) $F(t) = f(\underbrace{t^2}_{u}, \underbrace{\ln t}_{v}),\ t>0$

$$\frac{\partial u}{\partial t} = 2t \qquad \frac{\partial v}{\partial t} = \frac{1}{t}$$

$$\frac{\partial F}{\partial t} = \frac{\partial}{\partial t}\left(f(t^2,\ln t)\right) = \frac{\partial f}{\partial u}\cdot\frac{\partial u}{\partial t} + \frac{\partial f}{\partial v}\cdot\frac{\partial v}{\partial t} = \frac{\partial f}{\partial u}\cdot 2t + \frac{\partial f}{\partial v}\cdot\frac{1}{t}$$

$$\Rightarrow dF = \left(\frac{\partial f}{\partial u}\cdot 2t + \frac{\partial f}{\partial v}\cdot\frac{1}{t}\right)dt$$

$$\frac{\partial^2 F}{\partial t^2} = \frac{\partial}{\partial t}\left(\frac{\partial f}{\partial u}\cdot 2t + \frac{\partial f}{\partial v}\cdot\frac{1}{t}\right) = 2\cdot\frac{\partial f}{\partial u} + \frac{\partial^2 f}{\partial u^2}\cdot 4t^2 - \frac{\partial f}{\partial v}\cdot\frac{1}{t} + \frac{\partial^2 f}{\partial v^2}\cdot\frac{1}{t^2} +$$

$$\frac{\partial^2 f}{\partial u\,\partial v}\cdot 2 + \frac{\partial^2 f}{\partial v\,\partial u}\cdot 2 \Rightarrow d^2F = \left(4t^2\frac{\partial^2 f}{\partial u^2} + 4\frac{\partial^2 f}{\partial u\,\partial v} + \frac{1}{t^2}\frac{\partial^2 f}{\partial v^2} + 2\frac{\partial f}{\partial u} - \frac{1}{t^2}\frac{\partial f}{\partial v}\right)dt^2$$

b) $G(x,y) = g(\underbrace{x^2}_{u}, \underbrace{\frac{x}{y}}_{v}), y \neq 0$

$\frac{\partial u}{\partial x} = 2x \quad \frac{\partial u}{\partial y} = 0 \quad \frac{\partial v}{\partial x} = \frac{1}{y} \quad \frac{\partial v}{\partial y} = -\frac{x}{y^2}$

$\frac{\partial G}{\partial x} = \frac{\partial}{\partial x}\left(g(x^2, \frac{x}{y})\right) = \frac{\partial g}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial g}{\partial v} \cdot \frac{\partial v}{\partial x} = 2x \cdot \frac{\partial g}{\partial u} + \frac{1}{y} \cdot \frac{\partial g}{\partial v}$

$\frac{\partial G}{\partial y} = \frac{\partial}{\partial y}\left(g(x^2, \frac{x}{y})\right) = \frac{\partial g}{\partial v} \cdot \frac{\partial v}{\partial y} = -\frac{\partial g}{\partial v} \cdot \frac{x}{y^2}$

$dG = \left(2x \frac{\partial g}{\partial u} + \frac{1}{y} \cdot \frac{\partial g}{\partial v}\right) dx - \frac{\partial g}{\partial v} \cdot \frac{x}{y^2} dy$

$\frac{\partial^2 G}{\partial x^2} = \frac{\partial}{\partial x}\left(2x \frac{\partial g}{\partial u} + \frac{1}{y} \frac{\partial g}{\partial v}\right) = 2 \frac{\partial g}{\partial u} + \frac{\partial^2 g}{\partial u^2} \cdot 4x^2 + \frac{\partial^2 g}{\partial v \partial u} \cdot \frac{2x}{y} +$

$\frac{1}{y^2} \cdot \frac{\partial^2 g}{\partial v^2} + \frac{2x}{y} \frac{\partial^2 g}{\partial u \partial v} = 2 \frac{\partial g}{\partial u} + \frac{\partial^2 g}{\partial u^2} \cdot 4x^2 + \frac{4x}{y} \frac{\partial^2 g}{\partial u \partial v} + \frac{1}{y^2} \frac{\partial^2 g}{\partial v^2}$

$\frac{\partial^2 G}{\partial x \partial y} = \frac{\partial}{\partial x}\left(-\frac{\partial g}{\partial v} \cdot \frac{x}{y^2}\right) = -\frac{\partial g}{\partial v} \cdot \frac{1}{y^2} - \frac{\partial^2 g}{\partial u \partial v} \cdot \frac{2x^2}{y^2} - \frac{\partial^2 g}{\partial v^2} \cdot \frac{x}{y^3}$

$\frac{\partial^2 G}{\partial y^2} = \frac{\partial}{\partial y}\left(-\frac{\partial g}{\partial v} \cdot \frac{x}{y^2}\right) = \frac{\partial g}{\partial v} \cdot \frac{2x}{y^3} + \frac{\partial^2 g}{\partial v^2} \cdot \frac{x^2}{y^4}$

$\Rightarrow d^2G = \left(4x^2 \frac{\partial^2 g}{\partial u^2} + \frac{4x}{y} \frac{\partial^2 g}{\partial u \partial v} + \frac{1}{y^2} \frac{\partial^2 g}{\partial v^2} + 2 \frac{\partial g}{\partial u}\right) dx^2 -$

$2\left(\frac{\partial g}{\partial v} \cdot \frac{1}{y^2} + \frac{\partial^2 g}{\partial u \partial v} \cdot \frac{2x^2}{y^2} + \frac{\partial^2 g}{\partial v^2} \cdot \frac{x}{y^3}\right) dx dy + \left(\frac{\partial g}{\partial v} \cdot \frac{2x}{y^3} + \frac{\partial^2 g}{\partial v^2} \cdot \frac{x^2}{y^4}\right) dy^2$

c) $H(x,y,z) = h(\underbrace{x+y+z}_{u}, \underbrace{x^2+y^2+z^2}_{v})$

$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = \frac{\partial u}{\partial z} = 1 \quad \frac{\partial v}{\partial x} = 2x \quad \frac{\partial v}{\partial y} = 2y \quad \frac{\partial v}{\partial z} = 2z$

Diferențiala de ordinul I

$\frac{\partial f}{\partial x} = \frac{\partial h}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial h}{\partial v} \cdot \frac{\partial v}{\partial x} = \frac{\partial h}{\partial u} + \frac{\partial h}{\partial v} \cdot 2x$

$\frac{\partial f}{\partial y} = \frac{\partial h}{\partial u} \cdot \frac{\partial u}{\partial y} + \frac{\partial h}{\partial v} \cdot \frac{\partial v}{\partial y} = \frac{\partial h}{\partial u} + \frac{\partial h}{\partial v} \cdot 2y$

$\frac{\partial f}{\partial z} = \frac{\partial h}{\partial u} \cdot \frac{\partial u}{\partial z} + \frac{\partial h}{\partial v} \cdot \frac{\partial v}{\partial z} = \frac{\partial h}{\partial u} + \frac{\partial h}{\partial v} \cdot 2z \Rightarrow$

$df = \frac{\partial h}{\partial u}(dx+dy+dz) + 2 \frac{\partial h}{\partial v}(x dx + y dy + z dz)$

Diferențiala de ordinul al II-lea

$\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 h}{\partial u^2} \cdot \frac{\partial u}{\partial x} + \frac{\partial^2 h}{\partial v \partial u} \cdot \frac{\partial v}{\partial x} + 2 \frac{\partial h}{\partial v} + 2x \frac{\partial h}{\partial v^2} \cdot \frac{\partial v}{\partial x} + 2x \cdot \frac{\partial h}{\partial u \partial v} \cdot \frac{\partial u}{\partial x} =$

$= \frac{\partial^2 h}{\partial u^2} + 4x \frac{\partial^2 h}{\partial u \partial v} + 4x^2 \frac{\partial^2 h}{\partial v^2} + 2 \frac{\partial h}{\partial v}$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 h}{\partial u^2} + 4y\frac{\partial^2 h}{\partial u \partial v} + 4y^2\frac{\partial^2 h}{\partial v^2} + 2\frac{\partial h}{\partial v}$$

$$\frac{\partial^2 f}{\partial z^2} = \frac{\partial^2 h}{\partial u^2} + 4z\frac{\partial^2 h}{\partial u \partial v} + 4z^2\frac{\partial^2 h}{\partial v^2} + 2\frac{\partial h}{\partial v}$$

$$\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial}{\partial x}\left(\frac{\partial h}{\partial u} + \frac{\partial h}{\partial v}\cdot 2y\right) = \frac{\partial^2 h}{\partial u^2}\cdot\frac{\partial u}{\partial x} + \frac{\partial^2 h}{\partial v \partial u}\cdot\frac{\partial v}{\partial x} +$$

$$\frac{\partial^2 h}{\partial u \partial v}\cdot\frac{\partial u}{\partial x}\cdot 2y + \frac{\partial^2 h}{\partial v^2}\cdot\frac{\partial v}{\partial x}\cdot 2y = \frac{\partial^2 h}{\partial u^2} + \frac{\partial^2 h}{\partial u \partial v}(2x+2y) + \frac{\partial^2 h}{\partial v^2}\cdot 4xy$$

$$\frac{\partial^2 f}{\partial y \partial z} = \frac{\partial^2 h}{\partial u^2} + 2\frac{\partial^2 h}{\partial u \partial v}(y+z) + \frac{\partial^2 h}{\partial v^2}\,4yz$$

$$\frac{\partial^2 f}{\partial z \partial x} = \frac{\partial^2 h}{\partial u^2} + 2\frac{\partial^2 h}{\partial u \partial v}(z+x) + \frac{\partial^2 h}{\partial v^2}\,4zx$$

$$d^2 f = \frac{\partial^2 h}{\partial u^2}(dx^2 + dy^2 + dz^2 + 2dxdy + 2dydz + 2dzdx) + \frac{\partial^2 h}{\partial u \partial v}(4x\,dx^2 +$$

$$4y\,dy^2 + 4z\,dz^2 + 4(x+y)\,dxdy + 4(y+z)\,dydz + 4(z+x)\,dzdx) + \frac{\partial^2 h}{\partial v^2}\cdot$$

$$(4x^2dx^2 + 4y^2dy^2 + 4z^2dz^2 + 4xy\,dxdy + 4yz\,dydz + 4zx\,dzdx) +$$

$$\frac{\partial h}{\partial u}(2dx^2 + 2dy^2 + 2dz^2) = \frac{\partial^2 h}{\partial u^2}(dx+dy+dz)^2 + 4\frac{\partial^2 h}{\partial v^2}(x\,dx + y\,dy + z\,dz)^2 +$$

$$4\frac{\partial^2 h}{\partial u \partial v}(x\,dx^2 + y\,dy^2 + z\,dz^2 + xy\,dxdy + yz\,dydz + zx\,dzdx) + 2\frac{\partial h}{\partial v}(dx^2 +$$

$$dy^2 + dz^2).$$

d) $f(x,y,z) = l(\underbrace{x^y}_{u}, \underbrace{y^z}_{v}, \underbrace{z^x}_{w})$, $x,y,z>0$

$$\frac{\partial u}{\partial x} = y\cdot z^{y-1} \qquad \frac{\partial v}{\partial x} = 0 \qquad \frac{\partial w}{\partial x} = \ln z\cdot z^x$$

$$\frac{\partial u}{\partial y} = \ln x\cdot x^y \qquad \frac{\partial v}{\partial y} = z\cdot y^{z-1} \qquad \frac{\partial w}{\partial y} = 0$$

$$\frac{\partial u}{\partial z} = 0 \qquad \frac{\partial v}{\partial z} = \ln y\cdot y^z \qquad \frac{\partial w}{\partial z} = x\cdot z^{x-1}$$

Diferențiala de ordinul I

$$\frac{\partial f}{\partial x} = \frac{\partial l}{\partial u}\cdot\frac{\partial u}{\partial x} + \frac{\partial l}{\partial v}\cdot\frac{\partial v}{\partial x} + \frac{\partial l}{\partial w}\cdot\frac{\partial w}{\partial x} = \frac{\partial l}{\partial u}\cdot y\cdot x^{y-1} + \frac{\partial l}{\partial w}\cdot\ln z\cdot z^x$$

$$\frac{\partial f}{\partial y} = \frac{\partial l}{\partial u}\cdot\ln x\cdot x^y + \frac{\partial l}{\partial v}\cdot z\cdot y^{z-1}$$

$$\frac{\partial f}{\partial z} = \frac{\partial l}{\partial v}\cdot\ln y\cdot y^z + \frac{\partial l}{\partial w}\cdot x\cdot z^{x-1}$$

$$df = \frac{\partial l}{\partial u}x^{y-1}(y\,dx + x\ln x\,dy) + \frac{\partial l}{\partial v}\cdot y^{z-1}(z\,dy + y\ln y\,dz) +$$

$$\frac{\partial l}{\partial w}\cdot z^{x-1}\cdot(x\,dz + z\ln z\,dx)$$

(29.)

$$f(x,y)=\begin{cases}\dfrac{1-\cos(x^2+y^2-1)}{x^2+y^2-1}, & \text{dacă } x^2+y^2\neq 1\\ 0, & x^2+y^2=1\end{cases}$$

Funcția este evident diferențiabilă în punctele $(x_0,y_0)\in\mathbb{R}^2$ unde $x_0^2+y_0^2\neq 1$.

Să studiem diferențiabilitatea funcției în punctele $(x,y)\in\mathbb{R}^2$ unde $x^2+y^2=1$.

Fie $x_0,y_0\in\mathbb{R}$ a.î. $x_0^2+y_0^2=1$.

$$\frac{\partial f}{\partial x}(x_0,y_0)=\lim_{x\to x_0}\frac{f(x,y_0)-f(x_0,y_0)}{x-x_0}=\lim_{x\to x_0}\frac{\dfrac{1-\cos(x^2+y_0^2-1)}{x^2+y_0^2-1}}{x-x_0}=$$

$$=\lim_{x\to x_0}\frac{\dfrac{2\sin^2\left(\frac{x^2+y_0^2-1}{2}\right)}{x^2+y_0^2-1}}{x-x_0}=\lim_{x\to x_0}\frac{\sin^2\left(\frac{x^2+y_0^2-1}{2}\right)}{\left(\frac{x^2+y_0^2-1}{2}\right)^2}\cdot\frac{1}{2}\cdot\frac{(x^2+y_0^2-1)}{x-x_0}=$$

$$=\lim_{x\to x_0}\frac{1}{2}\cdot\frac{x^2+y_0^2-1}{x-x_0}=\lim_{x\to x_0}\frac{1}{2}\cdot\frac{x^2+y_0^2-x_0^2-y_0^2}{x-x_0}=\lim_{x\to x_0}\frac{1}{2}\cdot\frac{(x-x_0)(x+x_0)}{x-x_0}$$

$$=\lim_{x\to x_0}\frac{1}{2}\cdot(x+x_0)=x_0$$

Analog obținem că $\frac{\partial f}{\partial y}(x_0,y_0)=y_0$

$$\lim_{(x,y)\to(x_0,y_0)}\frac{f(x,y)-f(x_0,y_0)-\frac{\partial f}{\partial x}(x_0,y_0)(x-x_0)-\frac{\partial f}{\partial y}(x_0,y_0)\cdot(y-y_0)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}=$$

$$=\lim_{(x,y)\to(x_0,y_0)}\frac{f(x,y)-x_0(x-x_0)-y_0(y-y_0)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}=\lim_{(x,y)\to(x_0,y_0)}\frac{\frac{1-\cos(x^2+y^2-1)}{x^2+y^2-1}-x_0(x-x_0)-y_0(y-y_0)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}$$

$$=\lim_{(x,y)\to(x_0,y_0)}\frac{\frac{2\sin^2(x^2+y^2-1)}{(x^2+y^2-1)^2}\cdot\frac{1}{2}\cdot(x^2+y^2-1)-x_0(x-x_0)-y_0(y-y_0)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}=$$

$$=\lim_{(x,y)\to(x_0,y_0)}\frac{x^2+y^2-1-2(x_0x+y_0y-x_0^2-y_0^2)}{\sqrt{(x-x_0)^2+(y-y_0)^2}}=\lim_{(x,y)\to(x_0,y_0)}\frac{x^2+y^2+1-2(x_0x+y_0y)}{2\sqrt{(x-x_0)^2+(y-y_0)^2}}$$

$$=\lim_{(x,y)\to(x_0,y_0)}\frac{(x-x_0)^2+(y-y_0)^2}{\sqrt{(x-x_0)^2+(y-y_0)^2}}=\frac{1}{2}\lim_{(x,y)\to(x_0,y_0)}\sqrt{(x-x_0)^2+(y-y_0)^2}=0\Rightarrow$$

f este diferențiabilă în (x_0,y_0).

Cum (x_0,y_0) au fost arbitrar alese $\Rightarrow$ f este diferențiabilă pe $\mathbb{R}^2$

(30) **Matricea jacobiană**

$F: \mathbb{R}^2 \to \mathbb{R}^2$, $F(x,y) = (\overbrace{x+y^2}^{u}, \overbrace{xe^y}^{v})$

$$J_F = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix} = \begin{pmatrix} 1 & 2y \\ e^y & xe^y \end{pmatrix}$$

(31) **Matricea jacobiană**

$F: \mathbb{R}^3 \setminus \{yz \le 0\} \to \mathbb{R}^2$ $F(x,y,z) = (\overbrace{xy^2}^{u}, \overbrace{y\ln z}^{v})$

$$J_F = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \end{pmatrix} = \begin{pmatrix} y^2 & 2xy & 0 \\ 0 & \ln z & \frac{y}{z} \end{pmatrix}$$

(32) $f(x,y,z) = \frac{xy}{z}\ln x + x\varphi(\overbrace{\frac{y}{x}}^{u}, \overbrace{\frac{z}{x}}^{v})$, $x>0$, $z\neq 0$

$\frac{\partial u}{\partial x} = \frac{-y}{x^2}$ $\frac{\partial u}{\partial y} = \frac{1}{x}$ $\frac{\partial u}{\partial z} = 0$

$\frac{\partial v}{\partial x} = \frac{-z}{x^2}$ $\frac{\partial v}{\partial y} = 0$ $\frac{\partial v}{\partial z} = \frac{1}{x}$

$\frac{\partial f}{\partial x} = \frac{y}{z}\ln x + \frac{y}{z} + \varphi(\frac{y}{x}, \frac{z}{x}) + x\cdot\frac{\partial \varphi}{\partial u}\cdot\frac{\partial u}{\partial x} + z\cdot\frac{\partial \varphi}{\partial v}\cdot\frac{\partial v}{\partial x} = \frac{y}{z}(\ln x + 1) -$

$\frac{\partial \varphi}{\partial u}\cdot\frac{y}{x} - \frac{\partial \varphi}{\partial v}\cdot\frac{z}{x} + \varphi(\frac{y}{x}, \frac{z}{x})$

$\frac{\partial f}{\partial y} = \frac{x}{z}\ln x + z\cdot\frac{\partial \varphi}{\partial u}\cdot\frac{\partial u}{\partial y} + z\cdot\frac{\partial \varphi}{\partial v}\cdot\frac{\partial v}{\partial y} = \frac{x}{z}\ln x + \frac{\partial \varphi}{\partial u}$

$\frac{\partial f}{\partial z} = -\frac{xy}{z^2}\ln x + x\cdot\frac{\partial \varphi}{\partial v}\cdot\frac{1}{x} = \frac{-xy}{z^2}\cdot\ln x + \frac{\partial \varphi}{\partial v} \Rightarrow$

$x\frac{\partial f}{\partial x} + y\frac{\partial f}{\partial y} + z\frac{\partial f}{\partial z} - \frac{xy}{z} - f(x,y,z) = \cancel{\frac{xy}{z}\ln x} + \cancel{\frac{xy}{z}} - \cancel{y\cdot\frac{\partial \varphi}{\partial u}}$

$+ x\varphi(\frac{y}{x}, \frac{z}{x}) - \cancel{z\frac{\partial \varphi}{\partial u}} + \cancel{\frac{xy}{z}\ln x} + \cancel{y\cdot\frac{\partial \varphi}{\partial u}} - \cancel{\frac{xy}{z}\ln x} + \cancel{\frac{\partial \varphi}{\partial v}\cdot z} -$

$\cancel{\frac{xy}{z}} - \cancel{\frac{xy}{z}\ln x} - x\varphi(\frac{y}{x}, \frac{z}{x}) = 0$

(33) $f(x,y) = y^n\cdot\varphi(\overbrace{\frac{x}{y}}^{u}) + y^{1-n}\psi(\overbrace{\frac{x}{y}}^{u})$

$\frac{\partial u}{\partial x} = \frac{1}{y}$ $\frac{\partial u}{\partial y} = \frac{-x}{y^2}$

I $\begin{cases} \frac{\partial f}{\partial x} = y^{n-1}\frac{\partial \varphi}{\partial u} + y^{-n}\cdot\frac{\partial \psi}{\partial u} \\ \frac{\partial f}{\partial y} = n\cdot y^{n-1}\cdot\varphi(\frac{x}{y}) + (1-n)\cdot y^{-n}\psi(\frac{x}{y}) - \frac{\partial \varphi}{\partial u}\cdot x\cdot y^{n-2} - \frac{\partial \psi}{\partial u}\cdot x\cdot y^{-n-1} \end{cases}$

Derivate partiale de ord 2

$$\frac{\partial^2 f}{\partial x^2} = y^{u-2}\cdot\frac{\partial^2\varphi}{\partial u^2} + y^{-u-1}\cdot\frac{\partial^2\psi}{\partial u^2}$$

$$\frac{\partial^2 f}{\partial x\partial y} = n\cdot y^{u-2}\cdot\frac{\partial\varphi}{\partial u} + (1-n)\cdot y^{-u-1}\frac{\partial\psi}{\partial u} - \frac{\partial\varphi}{\partial u}\cdot y^{u-2} - \frac{\partial\psi}{\partial u}\cdot y^{-u-1} - \frac{\partial^2\varphi}{\partial u^2}\cdot x\cdot y^{u-3} -$$

$$\frac{\partial^2\psi}{\partial u^2}\cdot x y^{-u-2} = (u-1)\cdot y^{u-2}\cdot\frac{\partial\varphi}{\partial u} - u\cdot y^{-u-1}\frac{\partial\psi}{\partial u} - \frac{\partial^2\varphi}{\partial u^2}\cdot x y^{u-3} -$$

$$\frac{\partial^2\psi}{\partial u^2}\cdot x y^{-u-2}$$

$$\frac{\partial^2 f}{\partial y^2} = u(u-1)y^{u-2}\varphi\left(\frac{x}{y}\right) - u(1-u)y^{-u-1}\psi\left(\frac{x}{y}\right) - (u-2)x y^{u-3}\frac{\partial\varphi}{\partial u} +$$

$$x(u+1)y^{-u-2}\frac{\partial\psi}{\partial u} + - \frac{\partial\varphi}{\partial u}nx\cdot y^{u-3} - (1-u)y^{-u-2}\cdot x\cdot\frac{\partial\psi}{\partial u} + \frac{\partial^2\varphi}{\partial u^2}x^2 y^{u-4} +$$

$$\frac{\partial^2\psi}{\partial u^2}\cdot x^2 y^{-u-3} = u(u-1)y^{u-2}\varphi\left(\frac{x}{y}\right) + u(u-1)y^{-u-1}\psi\left(\frac{x}{y}\right) - (2u-2)x y^{u-3}\frac{\partial\varphi}{\partial u} +$$

$$2nx y^{-u-2}\frac{\partial\psi}{\partial u} + \frac{\partial^2\varphi}{\partial u^2}x^2 y^{u-4} + \frac{\partial^2\psi}{\partial u^2}x^2 y^{-u-3}$$

$$\Rightarrow x^2\frac{\partial^2 f}{\partial x^2} + 2xy\frac{\partial^2 f}{\partial x\partial y} + y^2\frac{\partial^2 f}{\partial y^2} = x^2 y^{u-2}\frac{\partial^2\varphi}{\partial u^2} + x^2 y^{-u-1}\frac{\partial^2\psi}{\partial u^2} +$$

$$2x(u-1)y^{u-1}\frac{\partial\varphi}{\partial u} - 2nxy^{-u}\frac{\partial\psi}{\partial u} - 2x^2 y^{u-2}\frac{\partial^2\varphi}{\partial u^2} - 2x^2 y^{-u-1}\frac{\partial^2\psi}{\partial u^2} +$$

$$u(u-1)y^u\cdot\varphi\left(\frac{x}{y}\right) + u(u-1)y^{-u+1}\psi\left(\frac{x}{y}\right) - (2u-2)\cdot x y^{u-1}\cdot\frac{\partial\varphi}{\partial u} +$$

$$2n\cdot x y^{-u}\cdot\frac{\partial\psi}{\partial u} + \frac{\partial^2\varphi}{\partial u^2}x^2 y^{u-2} + \frac{\partial^2\psi}{\partial u^2}x^2 y^{-u-1} = u(u-1)y^u\varphi\left(\frac{x}{y}\right) +$$

$$u(u-1)y^{1-u}\psi\left(\frac{x}{y}\right) = u(u+1)f.$$

(34.) Diferențiabile de ord. I și II

$$f(x,y,z) = xyz$$

(I)
$$\left.\begin{aligned}\frac{\partial f}{\partial x} &= yz\\ \frac{\partial f}{\partial y} &= xz\\ \frac{\partial f}{\partial z} &= xy\end{aligned}\right\} \Rightarrow df = yz\,dx + xz\,dy + xy\,dz$$

(II)
$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial y^2} = \frac{\partial^2 f}{\partial z^2} = 0$$

$$\frac{\partial f}{\partial x\partial y} = z \qquad \frac{\partial f}{\partial y\partial z} = x \qquad \frac{\partial f}{\partial z\partial x} = y$$

$$d^2 f = 2z\,dz\,dy + 2x\,dy\,dz + 2y\,dz\,dx$$